

D803

Natural Language Processing

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

An appliance maker with 9 factories built a review sentiment classifier. Its pipeline lowercases text and removes words on a standard stopwords list, which includes “not” and “no.” Reviews such as “not good at all” are now labeled positive. What is the most likely cause?

- A. Stopword removal deleted the negations, leaving “good.”
- B. The model is too small and needs more layers.
- C. Lowercasing changed the meaning of the words.
- D. Reviews from different factories use different vocabulary.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A named entity recognizer for news articles starts missing many company and city names right after a new preprocessing step is added. Which step most likely caused it?

- A. Removing “not” and “no” as stopwords
- B. Lowercasing all text
- C. None; preprocessing never affects NER.
- D. Splitting text into tokens on spaces

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A legal-tech firm with 45 employees indexes 1,000 contracts. The term “indemnify” appears 3 times in a 100-word clause and appears in 10 of the 1,000 documents. Using $tf = \text{count} \div \text{length}$ and $idf = \log_{10}(N \div df)$, what is the term’s TF-IDF in the clause?

- A. 6
- B. 0.06
- C. 0.03
- D. -0.06

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

In a corpus of 50,000 documents, a term has $\text{idf} = \log_{10}(N \div \text{df}) = 3$. In how many documents does it appear?

- A.** 1,000
- B.** 50,000,000
- C.** About 16,667
- D.** 50

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A help-desk search tool uses bag-of-words matching. A query for “laptop won’t boot” fails to find the article “Notebook fails to start.” The company keeps 3,000 articles in an on-premises wiki. What is the best fix?

- A.** Add more words to the stopwords list.
- B.** Move the wiki to a cloud host.
- C.** Represent queries and articles with dense embeddings and rank by cosine similarity.
- D.** Switch to one-hot word vectors.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A hospital records team wants to pull every medication name and dose out of 80,000 discharge notes written over six years. Which way of framing the NLP task fits?

- A.** Keyword search for the word “medication.”
- B.** Token-level sequence labeling (named entity recognition) with tags for medication and dose spans.
- C.** Document classification with one label per note.
- D.** Sentiment analysis of each note.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A next-word model built on a simple RNN fails when a subject at the start of a 40-word sentence should decide the verb form at the end. It was trained on 2 GB of news from four countries. Which change most directly targets the failure?

- A. Switch to a bag-of-words model.
- B. Increase the vocabulary size.
- C. Add news from more countries.
- D. Replace the simple RNN with an LSTM.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Why does an LSTM handle long-range dependencies better than a simple RNN?

- A. Its gates control a cell state that is updated additively, so gradients fade less over many steps.
- B. It never suffers vanishing gradients at any length.
- C. It relates all tokens at once through self-attention.
- D. It removes the recurrent connection.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A team builds a transformer encoder from scratch for 12 languages, using 8 attention heads, but forgets to add positional encodings. The model gives the same representation for “dog bites man” and “man bites dog.” Why?

- A. Self-attention ignores order on its own; positional encodings supply it.
- B. Transformers need a recurrent layer to read order.
- C. The tokenizer lowercased the sentences.
- D. Eight heads are too few to learn word order.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

What mainly lets transformers train faster than RNNs on long sequences using GPUs?

- A. They read tokens strictly one at a time.
- B. They always have fewer parameters.
- C. Self-attention processes all positions in parallel instead of step by step.
- D. Attention cost always grows linearly with length.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A travel company compares two English-to-Spanish translation systems on the same test set with reference translations. System A scores BLEU 34 and System B scores 28. System B runs on cheaper hardware. What does A's higher score show?

- A. A is guaranteed to be more fluent and accurate.
- B. A finds more named entities.
- C. A's output overlaps more with the reference translations; human review should confirm quality.
- D. B is better because it costs less to run.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A startup has 2,000 labeled support tickets in 5 categories and a modest cloud budget. Its founders previously worked at a large bank. What is the best way to build a ticket classifier?

- A. Train a large transformer from scratch on the 2,000 tickets.
- B. Pick a model trained only on banking text, given the founders' background.
- C. Fine-tune a pretrained model but swap in a different tokenizer.
- D. Fine-tune a pretrained transformer, using that model's own tokenizer.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

An ~~appliance maker with 9 factories~~ built a review sentiment classifier. Its pipeline lowercases text and removes words on a standard stopwords list, which includes “not” and “no.” Reviews such as “not good at all” are now labeled positive. What is the most likely cause?

Target word: which preprocessing step broke sentiment **Decides:** Negation words are on the stopwords list.

A	Correct	Without “not,” the phrase reads as positive.
B	Adjacent category	No model can recover words deleted before training.
C	Confusion set	Case rarely changes sentiment; negation removal does.
D	Camouflage	The factory count does not decide the failure.

2. Answer B Concept 1 • transfer

A named entity recognizer for news articles starts missing many company and city names right after a new preprocessing step is added. Which step most likely caused it?

Target word: reverse: from NER failure to the step **Decides:** NER leans on capitalization cues.

A	Confusion set	Hurts sentiment, not entity names.
B	Correct	Removes the capital letters that signal proper nouns.
C	Absolute language	Preprocessing can remove the very features NER uses.
D	Adjacent category	Tokenization is required and does not erase entity cues.

3. Answer B Concept 2

A legal-tech firm ~~with 45 employees~~ indexes 1,000 contracts. The term “indemnify” appears 3 times in a 100-word clause and appears in 10 of the 1,000 documents. Using $tf = \text{count} \div \text{length}$ and $idf = \log_{10}(N \div df)$, what is the term’s TF-IDF in the clause?

Target word: TF-IDF value **Decides:** $tf = 0.03$; $idf = \log_{10}(100) = 2$.

A	Wrong stem number	Uses the raw count 3 instead of $3 \div 100$.
B	Correct	$0.03 \times 2 = 0.06$.
C	Dropped element	Term frequency only; drops the idf factor.
D	Inverted operation	Uses $\log_{10}(df \div N)$, flipping the ratio.

4. Answer D Concept 2 • transfer

In a corpus of 50,000 documents, a term has $idf = \log_{10}(N \div df) = 3$. In how many documents does it appear?

Target word: solve for document frequency **Decides:** $N \div df = 10^3 = 1,000$.

A	Confusion set	That is the ratio $N \div df$, not df .
B	Inverted operation	Multiplies N by 1,000 instead of dividing.
C	Dropped element	$50,000 \div 3$ skips undoing the log.
D	Correct	$50,000 \div 1,000 = 50$.

5. Answer C Concept 3

A help-desk search tool uses bag-of-words matching. A query for “laptop won’t boot” fails to find the article “Notebook fails to start.” ~~The company keeps 3,000 articles in an on-premises wiki.~~ What is the best fix?

Target word: fix for synonym mismatch **Decides:** Same meaning, no shared words.

A	Adjacent category	Removing words cannot create a match between synonyms.
B	Camouflage	Hosting does not decide matching quality.
C	Correct	Embeddings place synonyms like laptop and notebook close together.
D	Confusion set	One-hot vectors treat every word as unrelated, just like bag-of-words.

6. Answer B Concept 4

A hospital records team wants to pull every medication name and dose out of 80,000 discharge notes ~~written over six years.~~ Which way of framing the NLP task fits?

Target word: task formulation **Decides:** The goal is extracting spans inside text.

A	Dropped element	Finds the label word, not the actual drug names and doses.
B	Correct	NER marks which tokens belong to each entity.
C	Adjacent category	Gives one label per note, not the names inside it.
D	Confusion set	Measures tone, not entities.

7. Answer D Concept 5

A next-word model built on a simple RNN fails when a subject at the start of a 40-word sentence should decide the verb form at the end. ~~It was trained on 2 GB of news from four countries.~~ Which change most directly targets the failure?

Target word: fix for long-range dependency **Decides:** Information must survive many time steps.

A	Dropped element	Drops word order entirely.
B	Adjacent category	More words do not help memory across steps.
C	Camouflage	Data source does not fix the architecture’s memory limit.
D	Correct	Gated cell state carries information across long spans.

8. Answer A Concept 5 • transfer

Why does an LSTM handle long-range dependencies better than a simple RNN?

Target word: reverse: from the behavior to the mechanism **Decides:** Gates and a cell state.

A	Correct	This eases the vanishing-gradient problem.
B	Absolute language	It reduces the problem; very long spans still degrade.
C	Confusion set	That describes transformers.
D	Inverted operation	LSTMs are still recurrent.

9. Answer A Concept 6

A team builds a transformer encoder from scratch for 12 languages, using 8 attention heads, but forgets to add positional encodings. The model gives the same representation for “dog bites man” and “man bites dog.” Why?

Target word: why order is lost **Decides:** No positional encodings.

A	Correct	Without them, the same set of tokens looks the same in any order.
B	Adjacent category	Positional encodings, not recurrence, provide order.
C	Confusion set	Casing does not change the order problem.
D	Camouflage	Head count does not create order information.

10. Answer C Concept 6 · transfer

What mainly lets transformers train faster than RNNs on long sequences using GPUs?

Target word: reverse: from speed to cause **Decides:** Parallel processing of all positions.

A	Inverted operation	That is how RNNs work, and it is why they are slower.
B	Adjacent category	Transformers are often larger; speed comes from parallelism.
C	Correct	GPUs handle the parallel matrix operations efficiently.
D	Absolute language	Standard self-attention grows with the square of sequence length.

11. Answer C Concept 7

A travel company compares two English-to-Spanish translation systems on the same test set with reference translations. System A scores BLEU 34 and System B scores 28. ~~System B runs on cheaper hardware.~~ What does A's higher score show?

Target word: meaning of a BLEU difference **Decides:** BLEU measures n-gram overlap with references.

A	Absolute language	Valid translations can use different words and score lower.
B	Confusion set	BLEU is not an entity-recall measure.
C	Correct	BLEU is an overlap proxy, not a direct quality measure.
D	Camouflage	Cost does not decide what BLEU measures.

12. Answer D Concept 8

A startup has 2,000 labeled support tickets in 5 categories and a modest cloud budget. ~~Its founders previously worked at a large bank.~~ What is the best way to build a ticket classifier?

Target word: approach with little labeled data **Decides:** Only 2,000 labels.

A	Adjacent category	Far too little data to learn language from zero.
B	Camouflage	The founders' history does not decide the domain of the tickets.
C	Dropped element	Breaks the match between token IDs and learned embeddings.
D	Correct	Pretraining supplies language knowledge; fine-tuning adapts it cheaply.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	4	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Preprocessing must fit the task	Each cleanup step removes information: stopword lists often delete negations that sentiment needs, and lowercasing erases the capital letters NER relies on.	Applying one standard pipeline to every task.	A task that fails right after a “standard” cleanup step.
2	TF-IDF weighting	TF-IDF = term frequency × log(N ÷ document frequency), so words common across the corpus get low weight.	Using raw counts or forgetting the log.	A count, a document length, N, and a document frequency.
3	Dense embeddings capture similarity	Word and sentence embeddings place related meanings near each other, so cosine similarity matches synonyms that bag-of-words misses.	Expecting one-hot or bag-of-words vectors to match synonyms.	Same meaning, different words, and no match.
4	NER is sequence labeling	Named entity recognition tags spans of tokens (often with BIO labels); text classification assigns one label to a whole document.	Treating extraction of names as document classification.	“Pull out every name / drug / place” vs. “label each document.”
5	RNN vs. LSTM	LSTMs add gates and a cell state that carry information across many steps, easing the vanishing gradients that limit simple RNNs.	Fixing long-range failures with more vocabulary or data.	A word early in a long sentence decides something at the end.
6	Transformers and self-attention	Self-attention relates all tokens in parallel and has no built-in sense of order, so positional encodings are required.	Assuming attention knows word order, or that transformers read one token at a time.	Reordered sentences treated the same, or a question about training speed.
7	Machine translation and BLEU	Translation uses encoder-decoder models; BLEU scores n-gram overlap with reference translations and is a proxy, not proof of quality.	Reading a higher BLEU as a guarantee of better translations.	Two systems compared by BLEU on the same test set.
8	Fine-tuning pretrained models	With limited labeled data, fine-tune a pretrained model using its own tokenizer rather than training from scratch.	Training a large model from scratch or swapping in a different tokenizer.	A few thousand labeled examples and a modest budget.

Course scope (public description): Natural Language Processing offers an in-depth exploration of modern data-driven techniques for natural language processing (NLP). The course progresses from basic text preprocessing methods to advanced machine-learning (ML) models specifically tailored for NLP tasks. Topics include sentiment analysis, named entity recognition, machine translation, and the use of advanced neural network architectures such as recurrent neural networks (RNNs), long short-term memory (LSTM), and transformers. Industry-standard tools and frameworks are covered for practical applications and project-based learning. For successful completion of this course, learners should take AI and ML Foundations, Machine Learning for Computer Scientists, and Deep Learning prior to this course.